MOTIVIC SPECTRA

Abstract. In the last talk, we constructed ∞-category of spectra $\mathcal{S}\mathrm{p}$ by stabilizing the suspension functor on pointed spaces. The resulting category was then realized by other universal properties as well. After a brief review of this construction, we undertake the same procedure on pointed motivic spaces, which produces the Morel-Voevodsky's ∞-category of $\mathbb{A}^1$-motivic spectra. We end with some equivalent characterizations of $\mathbb{A}^1$-motivic spectra.

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1. Revisiting spectra

In the last talk, Thomas constructed the ∞-category of spectra in the following way.

Construction 1.1 (∞-category $\mathcal{S}\mathrm{p}$)**.** The whole construction of $\mathcal{S}\mathrm{p}$ is done inside the category of presentable ∞-categories with colimit preserving functors, denoted $\mathcal{P}\mathrm{r}^{\mathrm{L}}$. The ∞-category of Kan-complexes (spaces), denoted by $\mathcal{S}\mathrm{pc}$, is presentable (Example 5.5.1.8, [Lur09]). We begin with ∞-category Δ^0, denoted as $*$. Recall stabilization in homotopy theory means the following things, amongst others:

(1) should be cocomplete,
(2) should have a zero object,
(3) suspension should be invertible,
(4) should have a smash product compatible with all the above.

We first do the free cocompletion of $*$, that is, take the presheaf category $\mathcal{P}(*)$, which gives us back the category of spaces $\mathcal{S}\mathrm{pc}$. The benefit of this presentation is that we have the following universal property: for any cocomplete $\mathcal{D}$, the Yoneda embedding $* \hookrightarrow \mathcal{S}\mathrm{pc}$ gives the equivalence

$$\mathcal{F}\mathrm{un}^{\mathrm{colim}}(\mathcal{S}\mathrm{pc}, \mathcal{D}) \xrightarrow{\simeq} \mathcal{F}\mathrm{un}(*, \mathcal{D}) \simeq \mathcal{D}, \tag{1}$$

i.e. constructing colimit preserving functors out of $\mathcal{S}\mathrm{pc}$ to $\mathcal{D}$ is equivalent to choosing objects of $\mathcal{D}$. Note that the homotopy category of $\mathcal{S}\mathrm{pc}$ is the usual homotopy category of CW-complexes.

By going to we have already added all small colimits. But the initial and terminal objects are obviously different in $\mathcal{S}\mathrm{pc}$. We mitigate that by adding a base point: consider $\mathcal{S}\mathrm{pc}_*$ given by the slice ∞-category of spaces under $*$, i.e. pointed spaces

$$\mathcal{S}\mathrm{pc}_* := \mathcal{S}\mathrm{pc}_{*/}\,. \tag{2}$$

Clearly, this has a zero object given by $* \to *$. Next, we want suspension functor $\Sigma : \mathcal{S}\mathrm{pc}_* \to \mathcal{S}\mathrm{pc}_*$ to be invertible. This is handled by the general theory in the following way. Say a pointed ∞-category is *stable* if it has finite limits and colimits and a square is Cartesian if and only if it is co-Cartesian. In a pointed

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category, we may define for any morphism $f : c \to d$ its fiber (mapping path space) and cofiber (mapping cone) as follows:

$$\begin{array}{ccc} \mathrm{fib}(f) & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow \\ c & \xrightarrow[f]{} & d \end{array} \qquad \begin{array}{ccc} \mathrm{cofib}(f) & \longleftarrow & * \\ \uparrow & \ulcorner & \uparrow \\ d & \xleftarrow[f]{} & c \end{array}.$$

This leads to the following endofunctors on $\mathcal{S}\mathrm{pc}_*$:

$$\Sigma(c) := \mathrm{cofib}(c \to *) \quad \Omega(c) := \mathrm{fib}(* \to c).$$

As a pointed category is stable if and only if Σ and Ω form an equivalence (Theorem 1.1.2.14, [Lur17] or Remark 2.43, [Gal26]), therefore we ought to construct a pointed category where this happens. **It is at this place do we see one importance of working with ∞-categories; the ability to take colimits and limits of ∞-categories, without forgetting the homotopical information!**[1]

Indeed, an obvious way to make $\Sigma : \mathcal{S}\mathrm{pc}_* \to \mathcal{S}\mathrm{pc}_*$ invertible is to consider the colimit of the following diagram (compare this to the usual identification $\mathbb{Z}[1/p] = \varinjlim(\mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \to \dots)$):

$$\mathcal{S}\mathrm{pc}_* \xrightarrow{\Sigma} \mathcal{S}\mathrm{pc}_* \xrightarrow{\Sigma} \mathcal{S}\mathrm{pc}_* \to \dots. \tag{3.1}$$

This colimit exists in $\mathcal{P}\mathrm{r}^{\mathrm{L}}$ by Theorem 5.5.3.18, [Lur09]. On the other hand, by §5.5, [Lur09], we have that $\mathcal{P}\mathrm{r}^{\mathrm{L}\,\mathrm{op}} \simeq \mathcal{P}\mathrm{r}^{\mathrm{R}}$ and the colimits in $\mathcal{P}\mathrm{r}^{\mathrm{L}}$ are computed as limits of the corresponding right adjoints in $\mathcal{P}\mathrm{r}^{\mathrm{R}}$. This gives us, equivalent to above, a diagram whose limit is equivalent to the colimit of the above

$$\dots \to \mathcal{S}\mathrm{pc}_* \xrightarrow{\Omega} \mathcal{S}\mathrm{pc}_* \xrightarrow{\Omega} \mathcal{S}\mathrm{pc}_* \tag{3.2}$$

calculated in $\mathcal{C}\mathrm{AT}_\infty$. Denote both of them to be $\mathcal{S}\mathrm{p}$. Both descriptions carry universal properties for $\mathcal{S}\mathrm{p}$, but the latter one explains what are objects of $\mathcal{S}\mathrm{p}$ ought to be; any object in $\mathcal{S}\mathrm{p}$ is a family of objects $E_n \in \mathcal{S}\mathrm{pc}_*$ with identifications $E_n \xrightarrow{\simeq} \Omega E_{n+1}$, just the usual definition of an Ω-spectrum.

Thus $\mathcal{S}\mathrm{p}$ has adjunction

$$\Sigma^\infty : \mathcal{S}\mathrm{pc}_* \rightleftarrows \mathcal{S}\mathrm{p} : \Omega^\infty.$$

Let $\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}}$ be the subcateogry of $\mathcal{P}\mathrm{r}^{\mathrm{L}}$ of stable presentable ∞-categories and exact functors (preserving finite limits, equivalently finite colimits). By Corollary 1.4.2.27, [Lur17], the ∞-category $\mathcal{S}\mathrm{p}$ is a stable presentable ∞-category in $\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}}$. Thus, we have also obtained an invertible suspension by passing to $\mathcal{S}\mathrm{p}$. Observe that we have an universal property of $\mathcal{S}\mathrm{p}$ coming from the colimit presentation (see §1.4.4, [Lur17]): for any presentable stable ∞-category $\mathcal{D}$, evaluation at the sphere spectrum $\mathbb{S}$ gives an equivalence

$$\mathcal{F}\mathrm{un}^L(\mathcal{S}\mathrm{p}, \mathcal{D}) \xrightarrow{\simeq} \mathcal{D}. \tag{3.3}$$

Finally, to complete the construction, we need a symmetric monoidal structure on $\mathcal{S}\mathrm{p}$ which is compatible with limits and colimits and with smash product on $\mathcal{S}\mathrm{pc}_*$ via Σ^∞. Note that since $\Sigma X = X \wedge S^1$, therefore

$$\text{inverting } \Sigma \text{ can be seen as inverting } S^1. \tag{3.4}$$

This is Corollary 4.24, [Rob15]. Now consider $\mathcal{C}$ to be a symmetric monoidal ∞-category (see Definition 2.49, [Gal26]) and consider the ∞-category of commutative algebras $\mathcal{C}\mathrm{Alg}(\mathcal{C})$ in $\mathcal{C}$. Consider the symmetric monoidal structure on $\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}}$ given by Lurie tensor product (see §4.8.1, [Lur17]). Thus we can consider commutative algebras in $\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}}$, i.e. the ∞-category $\mathcal{C}\mathrm{Alg}(\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}})$. In this category, $\mathcal{S}\mathrm{p}$ admits a unique symmetric monoidal structure compatible with colimits such that $\Sigma^\infty : \mathcal{S}\mathrm{pc}_* \to \mathcal{S}\mathrm{p}$ is a symmetric monoidal functor and

$$\mathcal{S}\mathrm{p} \text{ is the initial object in } \mathcal{C}\mathrm{Alg}(\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}}). \tag{4}$$

[1]Same process can be done with model categories but at the cost of bringing more concepts. The benefit here is that we can deal limits and colimits of model categories at the same footing as usual limits and colimits in categories.

2. Constructing motivic spectra

Let $\mathbf{k}$ be an arbitrary base field and schemes are separated and finite-type over $\mathbf{k}$. Let $\mathcal{S}m_{\mathbf{k}}$ be the category of smooth schemes over $\mathbf{k}$ and give it the Nisnevich topology:

2.1. $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$ **and motivic spaces.** We first begin by passing to the ∞-category of $\mathcal{S}\mathrm{pc}_*$-valued presheaves over $\mathcal{S}\mathrm{m}_{\mathbf{k}}$, denoted $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$, so that we have all small colimits (each presheaf is a colimit of representables). Note that compared to cohomology theories for spaces, we have a based space valued presheaf(!) Now not all presheaves are going to behave geometrically nice. We dig out a class of them, via the following definition.

An $\mathbb{A}^1$-*motivic space* is then defined to be a a presheaf

$$F : \mathcal{S}\mathrm{m}_{\mathbf{k}}^{\mathrm{op}} \longrightarrow \mathcal{S}\mathrm{pc}_*$$

such that it satisfies

(1) (Homotopy invariance). We have $F(X) \xrightarrow{\simeq} F(X \times \mathbb{A}^1)$ for $X \in \mathcal{S}\mathrm{m}_{\mathbf{k}}$.
(2) (Nisnevich excision). F takes a Nisnevich square of $X \in \mathcal{S}\mathrm{m}_{\mathbf{k}}$ to a Cartesian square in $\mathcal{S}\mathrm{pc}_*$ with $F(\emptyset) = *$.

Recall that a Nisnevich square for $X \in \mathcal{S}\mathrm{m}_{\mathbf{k}}$ is a Cartesian square of the following type:

$$\begin{array}{ccc} W & \longrightarrow & V \\ \downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ U & \xrightarrow[j]{} & X \end{array} \tag{N}$$

where $j : U \to X$ is an open immersion and $f : V \to X$ is an étale morphism and the inclusion $(V - W)_{\mathrm{red}} \to (X - U)_{\mathrm{red}}$ induces an equivalence. This essentially captures the idea behind excisive pair (A, B) in classical topology, where the inclusion $(A, A \cap B) \hookrightarrow (X, B)$ induces an isomorphism in homology.

Example 2.1. Here is how a typical Nisnevich square might look like:

$$\begin{array}{ccc} \mathbb{A}^1 - \{0, \pm\lambda\} & \hookrightarrow & \mathbb{A}^1 - \{0, \lambda\} \\ \downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ \mathbb{A}^1 - \{\lambda^2\} & \xrightarrow[j]{} & \mathbb{A}^1 \end{array},$$

where f maps $x \mapsto x^2$. The map f is étale by the fact that Jacobian of f is

The category of motivic spaces is denoted

$$\mathcal{S}\mathrm{pc}_{\mathbf{k}} \subset \mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}}).$$

2.2. **Motivic localization.** Note that $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$ is a presentable ∞-category. By §5.5.4, [Lur09], we obtain a left adjoint to the inclusion $\mathcal{S}\mathrm{pc}_{\mathbf{k}} \hookrightarrow \mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$ which is called the *motivic localization*:

$$\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}}) \overset{L_{\mathbb{A}^1}}{\rightleftarrows} \mathcal{S}\mathrm{pc}_{\mathbf{k}}\,.$$

We ought to know what this $L_{\mathbb{A}^1}$ does to a presheaf. By §5.5.4, [Lur09], in-fact this is given by inverting precisely the morphisms fed into definition of a motivic space; the following theorem makes it precise.

Theorem 2.2. *Let $\mathcal{S}$ be the small set of maps in $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$ given by*

(1) *$y(X \times \mathbb{A}^1) \to y(X)$ for each $X \in \mathcal{S}\mathrm{m}_{\mathbf{k}}$,*
(2) *for any open Zariski cover $X = U \cup V$ and $W = U \cap V$, $y(U) \amalg_W y(V) \to y(X)$,*
(3) *$\emptyset \to y(\emptyset)$.*

Then $\mathcal{S}\mathrm{pc}_{\mathbf{k}} = S^{-1}\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$ and thus for any ∞-category $\mathcal{D}$, composition with $L_{\mathbb{A}^1}$ of any colimit preserving functor to an $\mathcal{S}\mathrm{pc}_{\mathbf{k}} \to \mathcal{D}$ gives an equivalence of $\mathcal{F}\mathrm{un}^L(\mathcal{S}\mathrm{pc}_{\mathbf{k}}, \mathcal{D})$ and functors in $\mathcal{F}\mathrm{un}^L(\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}}), \mathcal{D})$ which take morphisms in S to equivalences.

In particular, we have the composite

$$\mathcal{S}\mathrm{m}_{\mathbf{k}} \xrightarrow{y} \mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}}) \xrightarrow{L_{\mathbb{A}^1}} \mathcal{S}\mathrm{pc}_{\mathbf{k}}\,.$$

Remark 2.3 (Localization of Nisnevich squares)**.** Consider a Nisnevich square as in (N) in $\mathcal{S}\mathrm{m}_{\mathbf{k}}$ or equiuvalently, in $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$. The image of (N) under $L_{\mathbb{A}^1}$ is given by a coCartesian square. Indeed, it suffices to show that for any motivic space $F \in \mathcal{S}\mathrm{pc}_{\mathbf{k}}$, the functor

$$\mathrm{Map}_{\mathcal{S}\mathrm{pc}_{\mathbf{k}}}(L_{\mathbb{A}^1} y(-), F) : \mathcal{S}\mathrm{m}_{\mathbf{k}} \longrightarrow \mathcal{S}\mathrm{pc}_*$$

takes Nisnevich squares to Cartesian squares by adjunction $L_{\mathbb{A}^1}$ and Yoneda lemma, so that $L_{\mathbb{A}^1}$ maps it to a coCartesian square.

Remark 2.4 (Vector bundles in $\mathcal{S}\mathrm{pc}_{\mathbf{k}}$)**.** Consider a vector bundle $p : E \to B$ in $\mathcal{S}\mathrm{m}_{\mathbf{k}}$. This is an equivalence in $\mathcal{S}\mathrm{pc}_{\mathbf{k}}$ under $L_{\mathbb{A}^1} y$. Indeed, let U_i be a covering of B by open subschemes for which the restriction of p is the trivial bundle $U_i \times \mathbb{A}^n \to U_i$. Once again, we may check for equivalence by checking on an arbitrary motivic space F; $L_{\mathbb{A}^1} yp$ is an equivalence if and only if for any $F \in \mathcal{S}\mathrm{pc}_{\mathbf{k}}$, we have $\mathrm{Map}_{\mathcal{S}\mathrm{pc}_{\mathbf{k}}}(L_{\mathbb{A}^1} yB, F) \to \mathrm{Map}_{\mathcal{S}\mathrm{pc}_{\mathbf{k}}}(L_{\mathbb{A}^1} yE, F)$ is an equivalence. Again by Yoneda lemma and adjunction, we reduce to checking if

$$p^* : F(B) \longrightarrow F(E)$$

is an equivalence. Assume that $X = U \cap V$ where E is trivial on both. We thus have a Nisnevich square on the left below which on applied by F gives a Cartesian square on the right

$$\begin{array}{ccc} U \cap V & \longrightarrow & V \\ \downarrow & \lrcorner & \downarrow \\ U & \longrightarrow & B \end{array} \qquad \begin{array}{ccc} F(U \cap V) & \longleftarrow & F(V) \\ \uparrow & \ulcorner & \uparrow \\ F(U) & \longleftarrow & F(B) \end{array}.$$

We have a similar Nisnevich square for $E_U, E_{U\cap V}$ and E_V:

$$\begin{array}{ccc} E_{U\cap V} & \longrightarrow & E_V \\ \downarrow & \lrcorner & \downarrow \\ E_U & \longrightarrow & E \end{array} \qquad \begin{array}{ccc} F(E_{U\cap V}) & \longleftarrow & F(E_V) \\ \uparrow & \ulcorner & \uparrow \\ F(E_U) & \longleftarrow & F(E) \end{array}.$$

As the map $p^* : F(B) \to F(E)$ is induced from $p^* : F(V) \to F(E_V)$ and $p^* : F(U) \to F(E_U)$, both of which are equivalences by F being a motivic space. We then conclude by induction.

Remark 2.5 (Model structure on schemes)**.** The alternate way to build homotopy theory for schemes, before ∞-categories, would be to search for a model category structure on $\mathcal{S}\mathrm{m}_{\mathbf{k}}$. This is not possible as $\mathcal{S}\mathrm{m}_{\mathbf{k}}$ doesn't have enough colimits, so one passes to $\mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}})$, which all small colimits. Morel-Voevodsky gave a model structure on simplicial sheaves on $\mathcal{S}\mathrm{m}_{\mathbf{k}}$, after localization at Nisnevich and $\mathbb{A}^1$ locally.

2.3. **Stabilization.** Recall from Construction 1.1 that after constructing $\mathcal{S}\mathrm{pc}$, we construct the pointed category $\mathcal{S}\mathrm{pc}_*$ and then invert S^1 to obtain $\mathcal{S}\mathrm{p}$. In our situation, we may construct the pointed category quite easily as the slice of $\mathcal{S}\mathrm{pc}_{\mathbf{k}}$ under $* = \mathrm{Spec}\,(\mathbf{k})$. However, there are atleast two choices for spheres in $\mathcal{S}\mathrm{pc}_{\mathbf{k}}$: $\mathbb{G}_m = \mathrm{Spec}\left(\mathbf{k}[t, t^{-1}]\right)$ and $\mathbb{P}^1 = \mathrm{Proj}(\mathbf{k}[s, t])$. The question remains on which one to invert. It turns out that inverting $\mathbb{P}^1$ is enough.

Definition 2.6. The functor $\Sigma^\infty : (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_* \longrightarrow \mathcal{S}\mathrm{p}_{\mathbf{k}}$ obtained by inverting $(\mathbb{P}^1, \infty)$ is an initial map in $\mathcal{C}\mathrm{Alg}(\mathcal{P}\mathrm{r}^{\mathrm{L}})$ and $\mathcal{S}\mathrm{p}_{\mathbf{k}}$ is called the ∞-*category of motivic spectra*, or *stable* $\mathbb{A}^1$-*homotopy theory*.

Akin to case for $\mathcal{S}\mathrm{p}$, we have description of $\mathcal{S}\mathrm{p}_{\mathbf{k}}$ as a colimit in $\mathcal{P}\mathrm{r}^{\mathrm{L}}$ and limit in $\mathcal{C}\mathrm{AT}_\infty$. Indeed consider the functors

$$\Sigma : (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_* \longrightarrow (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_*$$

given by $- \wedge \mathbb{P}^1$ and the functor

$$\Omega : (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_* \longrightarrow (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_*$$

given by $\underline{\mathrm{Hom}}\left(\mathbb{P}^1, -\right)$, which is right adjoint to the smash with $\mathbb{P}^1$. It then follows from Corollary 4.24, [Rob15], that $\mathcal{S}\mathrm{p}_{\mathbf{k}}$ is also given by the following colimit in $\mathcal{P}\mathrm{r}^{\mathrm{L}}$

$$(\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}} \xrightarrow{\Sigma} (\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}} \xrightarrow{\Sigma} (\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}} \to \dots$$

and the following limit in $\mathcal{C}\mathrm{AT}_\infty$

$$\dots \to (\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}} \xrightarrow{\Omega} (\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}} \xrightarrow{\Omega} (\mathcal{S}\mathrm{pc}_*)_{\mathbf{k}}.$$

Once again, this gives the followin interpretations for the category of $\mathbb{A}^1$-motivic spectra:

(1) An object of $\mathcal{S}\mathrm{p}_{\mathbf{k}}$ is a sequence of motivic spaces E_n together with identifications $E_n \simeq \Omega E_{n+1}$. Untangling the definitions, an $\mathbb{A}^1$-motivic spectrum $\{E_n\}$ is a sequence of functors

$$E_n : \mathcal{S}\mathrm{m}_{\mathbf{k}}^{\mathrm{op}} \longrightarrow \mathcal{S}\mathrm{pc}_*$$

such that $E_n(X \times \mathbb{A}^1) \to E_n(X)$ is an equivalence, $E_n(\emptyset) = *$ and maps Nisnevich squares to pullback squares. Another name for an $\mathbb{A}^1$-motivic spectrum is a *Morel-Voevodsky cohomology theory*.

(2) By §1.4.4, [Lur17], we have the following universal property of $\mathcal{S}\mathrm{p}_{\mathbf{k}}$: for any $\mathcal{D} \in \mathcal{C}\mathrm{Alg}(\mathcal{P}\mathrm{r}^{\mathrm{L}}{}_{\mathrm{st}})$, the functor obtained by composing along the canonical map $\mathcal{S}\mathrm{m}_{\mathbf{k}} \longrightarrow \mathcal{S}\mathrm{p}_{\mathbf{k}}$ is a fully-faithful embedding

$$\mathcal{F}\mathrm{un}^{L,\otimes}(\mathcal{S}\mathrm{p}_{\mathbf{k}}, \mathcal{D}) \longrightarrow \mathcal{F}\mathrm{un}^{\otimes}(\mathcal{S}\mathrm{m}_{\mathbf{k}}, \mathcal{D})$$

whose essential image is the collection of monoidal functors $F : \mathcal{S}\mathrm{m}_{\mathbf{k}} \to \mathcal{D}$ satisfying
   (a) F sends $X \times \mathbb{A}^1 \to X$ to an equivalence in $\mathcal{D}$,
   (b) F maps a Nisnevich square to a coCartesian square in $\mathcal{D}$,
   (c) $F(\mathbb{P}^1)$ is an invertible object.

Proof. The whole process of constructing $\mathcal{S}\mathrm{p}_{\mathbf{k}}$ is given by:

$$\mathcal{S}\mathrm{m}_{\mathbf{k}} \xrightarrow{y} \mathcal{P}(\mathcal{S}\mathrm{m}_{\mathbf{k}}) \xrightarrow{L_{\mathbb{A}^1}} \mathcal{S}\mathrm{pc}_{\mathbf{k}} \xrightarrow{(-)_+} (\mathcal{S}\mathrm{pc}_{\mathbf{k}})_* \xrightarrow{\Sigma^\infty_{\mathbb{P}^1}} \mathcal{S}\mathrm{p}_{\mathbf{k}}\,.$$

The functors $(-)_+$ and $\Sigma^\infty_{\mathbb{P}^1}$ are colimit preserving, so that the result then follows from Theorem 2.2 and that $\Sigma^\infty_{\mathbb{P}^1}$ is a formal inversion at $\mathbb{P}^1$ (use its universal property). □

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